

System Security Project
Third Year, B.Tech in Information Technology
Veermata Jijabai Technological Institute (VJTI)

Team Members:

- Jay Ghorpade - 231080026
- Kajal Kumari - 231081034
- Sneha Singh - 231081067

Project Area: Web Application Security

Project Title: *Design and Implementation of a Web Application Firewall (WAF) for Real-Time Detection and Prevention of SQL Injection and Cross-Site Scripting (XSS) Attacks*

Abstract

Web applications serve as the primary interface for critical services including banking, health-care, and e-commerce, making them high-value targets for malicious actors. Among the most prevalent and damaging vulnerabilities identified by the OWASP Top 10 are SQL Injection (SQLi) and Cross-Site Scripting (XSS) attacks. SQLi exploits poorly sanitized database queries to allow unauthorized data access or manipulation, while XSS enables attackers to inject malicious scripts into web pages viewed by other users, compromising both client-side integrity and user data.

This project proposes the design and implementation of a lightweight Web Application Firewall (WAF) that operates as middleware within a web application stack. The system intercepts all incoming HTTP requests and applies a multi-layered detection mechanism comprising pattern-based signature matching, input sanitization filters, and a rule-based blacklist engine to identify and neutralize SQLi and XSS payloads in real time. A controlled vulnerable web application will be built to simulate realistic attack scenarios, against which the WAF will be tested and validated.

The implementation will demonstrate the system's ability to detect and block common attack vectors including tautology-based SQL injection, UNION-based injection, reflected XSS, and stored XSS, while preserving normal application functionality. Detection accuracy, false positive rate, and system response time will be evaluated as performance metrics. The project aims to provide both a functional security tool and a deeper understanding of how injection-based attacks operate and can be systematically countered at the application layer.

Base Research Paper

Title: An Approach to Detect and Prevent SQL Injection and XSS Vulnerability in the Web Application

Authors: Shekhar Disawal & Ugrasen Suman

Journal: International Journal of Engineering Trends and Technology (IJETT), Vol. 71, Issue 8, pp. 216–224, 2023

DOI: <https://doi.org/10.14445/22315381/IJETT-V71I8P219>

Justification: The selected paper proposes a Web Vulnerability Detection-Prevention Methodology (WV-DPM) utilizing the Rabin-Karp string-matching algorithm with a robust filter function. This approach is directly applicable to the proposed middleware WAF architecture. Furthermore, the paper provides a comprehensive comparative analysis of existing security techniques, providing a strong theoretical foundation for the project's implementation and performance evaluation.